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OmniKVQuant: KV Cache Quantization for Omni-LLMs

Temporal Key Drift and Heterogeneous Value Geometry Break Existing 2-Bit Methods on Multimodal Caches — Per-Window and Per-Modality Fixes Achieve Near-Lossless Quality

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Omni-modal large language models — models that jointly process audio, video, and text — are increasingly used in real-time deployments. Their KV cache grows proportionally to all modalities simultaneously: a 7B omni-model processing 10 minutes of audio-video accumulates ~ 30 GB of FP16 KV cache. 2-bit quantization would reduce this to 4 GB, enabling consumer deployment. But applying existing rotation-based quantization methods (designed for text-only LLMs) to multimodal caches cuts performance by 20 points or more. OmniKVQuant diagnoses why — two failure modes specific to multimodal caches — and fixes them with targeted, training-free interventions.

AT A GLANCE

Problem: Rotation-based KV cache quantization fails on omni-modal caches; no training-free solution exists.

Why it matters: Omni-LLMs for real-time audio-video require radical memory reduction; 2-bit is the practical target.

Method: Per-window key quantization (for temporal key drift); per-modality value rotation (for heterogeneous value geometry); fused Triton decode kernel.

Result: 2-bit KV cache on Qwen2.5-Omni and Qwen3-Omni with only 2.3-point degradation vs. 19–24 points for prior methods.

Evidence: 7 audio-visual benchmarks; ablations on window size, modality rotation, and kernel overhead.

KV Cache Memory in Omni-LLMs

The KV cache size scales as:

$$M_{KV} = 2 \cdot n_l \cdot n_h \cdot L \cdot d_h \cdot \text{prec}$$

For a 7B omni-model (32 layers, 32 heads, $d_h = 128$) processing 10 minutes of audio-video at 25 fps ($L \approx 15,000$), this is ≈ 30 GB at FP16 — far beyond consumer GPU memory.

Existing KV cache quantization for text-only LLMs uses rotation-based methods: apply an orthogonal matrix R to key/value vectors before quantization to eliminate outliers:

$$\tilde{V} = RV, \quad \hat{V} = Q_2(\tilde{V})$$

where Q_2 is 2-bit rounding. R is calibrated once on text calibration data. On text-only LLMs, this reduces memory to ≈ 4 GB with <1 point degradation. On omni-modal caches, applying the same R causes 19–24 point degradation.

Failure Mode 1: Temporal Key Drift

In a text-only LLM, the statistical distribution of key vectors is roughly stationary over the sequence. In an omni-modal model processing audio:

$$\mathbf{K}_t = W_K \cdot \mathbf{x}_t^{\text{audio}}$$

where $\mathbf{x}_t^{\text{audio}}$ is the audio embedding at time t . Audio embeddings have strong temporal autocorrelation — consecutive frames are highly similar — so the key distribution drifts systematically over time:

$$\|\mathbf{K}_t - \mathbf{K}_{t-1}\|_2 \ll \|\mathbf{K}_t\|_2$$

A quantization range $[r_{\min}, r_{\max}]$ calibrated at initialization clips new key values that drift outside this range, producing growing quantization error over longer audio streams.

Fix: per-window key quantization. Apply separate quantization ranges to each temporal window of size w :

$$r_{\min}^{(t)} = \min_{t' \in [t-w, t]} K_{j, t'}^{\min}, \quad r_{\max}^{(t)} = \max_{t' \in [t-w, t]} K_{j, t'}^{\max}$$

This adapts the quantization range locally, tracking drift without requiring global statistics. Window size $w = 64$ works well across all tested models.

Failure Mode 2: Heterogeneous Value Geometry

The optimal rotation R_m^* for modality m is the PCA rotation of the value covariance matrix Σ_m :

$$R_m^* \approx U_m^\top, \quad \Sigma_m = U_m \Lambda_m U_m^\top$$

These matrices are modality-specific:

- *Text:* Near-isotropic; ~ 30 directions with $>1\%$ variance.
- *Audio:* Concentrated; 90% variance in 5 low-frequency pitch/timbre directions.
- *Video:* High-rank anisotropic; large dynamic range in spatial-frequency directions.

Applying R_{text}^* to audio values leaves 85% of variance concentrated in 5 directions, producing massive outliers in the rotated space that 2-bit quantization cannot faithfully represent.

Fix: per-modality value rotation. Calibrate and store separate rotations $\{R_m^*\}$ per modality. At inference, detect the current modality from the token type embedding and apply the appropriate rotation:

$$\tilde{\mathbf{V}} = R_m^* \mathbf{V}, \quad \hat{\mathbf{V}} = Q_2(\tilde{\mathbf{V}})$$

Fused Triton Decode Kernel

Standard 2-bit implementations dequantize the cache to FP16 before attention, reconstructing the full dense cache and negating half the memory savings. OmniKVQuant provides a fused Triton kernel that dequantizes on the fly during attention computation, reading 2-bit integers per block and expanding to FP16 without storing the FP16 cache. Memory footprint remains at 2 bits throughout.

2-bit symmetric quantization:

$$\hat{v}_j = s \cdot \text{round}\left(\frac{v_j}{s}\right), \quad \text{round}(\cdot) \in \{-2, -1, 0, 1\}, \quad s = \frac{\max_j |v_j|}{2}$$

Experimental Results

Tested on Qwen2.5-Omni-7B and Qwen3-Omni-7B across 7 audio-visual benchmarks (LibriSpeech, MMAU, Video-MME, ActivityNet-QA, MSVD-QA, OmniBench, Evo-OMNIBench):

Method	Bits	Avg Score	Δ
Full FP16	16	72.4	—
TurboQuant	2	48.3	−24.1
KIVI	2	52.7	−19.7
OmniKVQuant	2	70.1	−2.3
OmniKVQuant	4	72.0	−0.4

OmniKVQuant reduces 2-bit degradation from 19–24 points (prior methods) to 2.3 points, and achieves near-lossless 4-bit quantization (0.4 points).

Ablations and Interpretation

Without per-window key quantization: Degradation grows from 2.3 to 11.4 points; confirms temporal key drift is the dominant failure mode.

Without per-modality value rotation: Degradation grows from 2.3 to 8.9 points; audio benchmarks are disproportionately affected (−18.3 on MMAU).

Kernel overhead: The fused Triton kernel adds 4% latency vs. FP16 attention while delivering $7.8\times$ memory reduction for 2-bit vs. FP16 caches.

Window size: $w = 64$ works well; smaller w slightly improves accuracy at the cost of more quantization parameter updates.

Reference

Suho Yoo, Hyunjong Ok, Jongmin Choi, Jihoo Jung, Joon Son Chung. **OmniKVQuant: KV Cache Quantization for Omni-LLMs.** arXiv:2609.11582, September 2026. <https://arxiv.org/abs/2609.11582>